

Sprint 6

Sprint 6 completed the remaining functional requirements of the Container Management System. The React frontend was fully integrated with the backend workflow engine, enabling complete lifecycle management through a responsive user interface. Dashboard analytics, Quick Scan, approval processing, and operational reporting were completed, providing users with improved visibility of container activity.

During the sprint, the lifecycle engine was refined to enforce business rules governing QA approval, lifecycle expiry, Supervisor approval, and lifecycle reset. End-to-end testing confirmed that containers could successfully progress through repeated production lifecycles while maintaining complete traceability and workflow validation.

The application is now functionally complete and entered the final testing, evaluation, and dissertation phase.

ID	User Story	Priority	Story Points	Status
PB22	Develop Container Management Interface	Must Have	5	Done
PB23	Implement Approval Workflow Interface	Must Have	8	Done
PB24	Enhance Dashboard with Operational Metrics	Must Have	5	Done
PB25	Implement Quick Search	Should Have	5	Done
PB26	Display Container Details	Must Have	5	Done
PB27	Implement Recent Activity Panel	Should Have	3	Done
PB-028	Integrate Operational Analytics	Should Have	5	Done

Definition of Done

- Container management interface completed.
- Approval workflow interface operational.
- Lifecycle expiry and Supervisor reset workflow fully validated
- Dashboard displays live operational statistics.
- Analytics integrated into the dashboard.
- Quick Scan implemented.
- Recent Activity panel operational.
- End-to-end workflow successfully tested.
- Code committed to GitHub.
- Documentation updated.

What Went Well

- Container management interface successfully completed.
- End-to-end approval workflow fully integrated.
- Dashboard analytics provided improved operational visibility.
- Quick Search simplified container lookup.
- Simplifying the workflow by removing the Observation module improved maintainability.
- Frontend and backend integration remained stable throughout development.

Challenges

- Synchronising frontend updates with backend workflow logic.
- Refining lifecycle expiry and Supervisor reset behaviour
- Validating business rules across repeated production lifecycles
- Maintaining consistent lifecycle reset behaviour after Supervisor approval.
- Refining the dashboard while preserving responsiveness.
- Balancing feature enhancements with project scope and deadlines.

At the conclusion of Sprint 6:

Backend workflow engine completed.

- React frontend completed.
- Authentication and role-based access control operational.
- Container lifecycle management completed.
- QA and Supervisor approval workflows completed.
- Dashboard and analytics implemented.
- Quick Scan implemented.
- Recent Activity implemented.
- End-to-end lifecycle and business rule validation complete
- Final dissertation and supporting documentation complete